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Final Assessment Presentation Slide

A Business Intelligence Solution for Early Detection and Prevention of Customer Churn in Telecom Networks

ANIEHE JOSHUA CHINONSO

Faculty of Engineering, University of Nigeria, Nsukka, Nigeria

25UNN_Joshua

Background & Global Context

~90%

of global telecommunication industries uses
Churn prediction to improve customers
retention

~30%

Churn is experienced in major industries annually

Customer Usage, Revenue,
and Tenure Will Be Among the Strongest
Churn Predictors

- Telecommunications industry experiences some of the highest churn rates among major industries, often exceeding 30% annually..
- Recent studies demonstrate that machine learning techniques such as LightGBM, XGBoost, Random Forest, and neural networks can effectively predict churn behavior by analyzing customer usage patterns, spending behavior, network experience, and demographic characteristics.
- The growth of mobile data services, 5G adoption, and digital communication platforms has increased customer expectations regarding service quality, network reliability, and pricing transparency.

Presentation outline

1. Business Problems
2. Research Goals & Objectives
3. Graphical Results of Exploratory Data Analysis
4. Recommendations and Insights
5. Overview of Machine Learning Model
6. Feature Importance
7. Model Prediction Process
8. Overall Recommendation
9. Quotation
10. Conclusion

Business Problems

Revenue Leakage and Declining Customer Value

The presence of revenue-related variables among the top predictors suggests that customer spending behavior is closely linked to churn. The company may be losing high-value customers or failing to maximize revenue from existing customers through upselling and personalized service plans.

Poor Customer Experience and Service Utilization

Changes in usage patterns and blocked data services may indicate customer dissatisfaction. Customers experiencing network congestion, poor data connectivity, or declining service engagement are more likely to switch providers.

High Customer Churn Due to Customer Lifecycle and Device Aging

Customers who have stayed for a long period or are using older devices may become dissatisfied with outdated technology, poor performance, or better offers from competitors. The company may not be effectively engaging customers throughout their lifecycle or encouraging timely device upgrades.

Geographic Variations in Customer Satisfaction and Network Performance

The importance of geographic variables suggests that churn rates differ significantly across regions. This may indicate inconsistent network quality, varying competitive pressures, or differences in customer service effectiveness across service areas.

GOAL: To develop a Business Intelligence solution that enables the early detection and prevention of customer churn in telecom networks through data-driven analysis and predictive modeling.

- 01 To identify and analyze the key factors contributing to customer churn in telecom networks, including customer tenure, device age, service usage, and revenue patterns.
- 02 To investigate geographical and customer-segment differences in churn behavior in order to identify high-risk regions and customer groups.
- 03 To evaluate the impact of customer spending behavior and revenue-related factors on customer retention and business profitability.
- 04 To provide actionable business intelligence recommendations that support customer retention, improve customer satisfaction, and reduce revenue loss due to churn.
- 05 To develop a predictive machine learning model capable of accurately identifying customers who are at risk of churning.

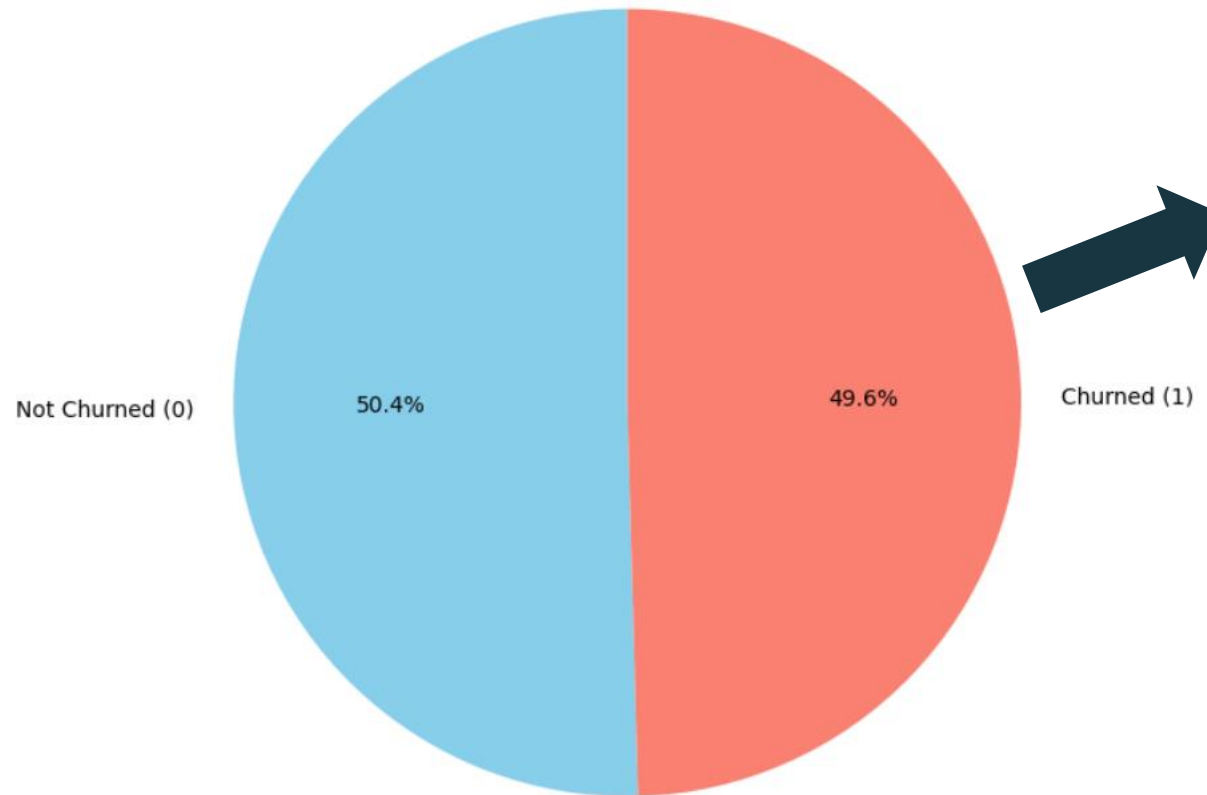
Graphical Results of the Exploratory Data Analysis

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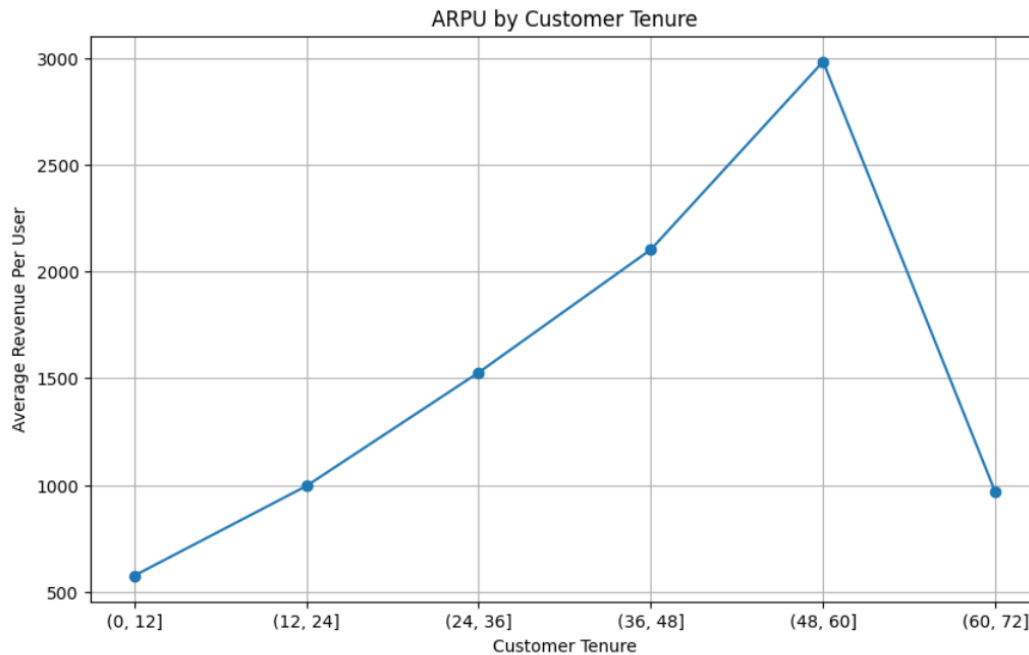
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Proportion of Churned vs. Non-Churned Customers



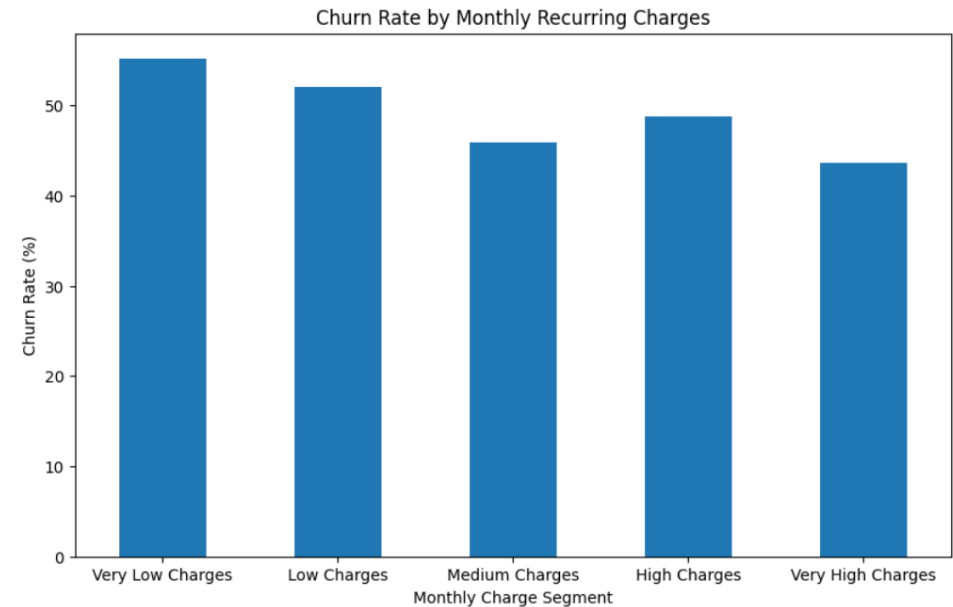
The company is experiencing a churn rate of approximately 50%, which represents a substantial loss of customers and poses a serious risk to revenue sustainability, customer lifetime value, and future business growth.

Graphical Evidence Supporting the First Business Problem (Revenue Leakage and Declining Customer Value)



Graphical Interpretation:

The analysis reveals a strong positive relationship between customer tenure and Average Revenue Per User (ARPU). Revenue contribution increases steadily as customers remain with the company, reaching its highest level among customers with 48–60 months of tenure. This suggests that long-term customers are significantly more valuable and should be the focus of retention strategies. However, the notable decline in ARPU among customers with more than 60 months of tenure warrants further investigation to identify potential causes such as plan downgrades, loyalty discounts, or changes in customer behavior.



Graphical Interpretation

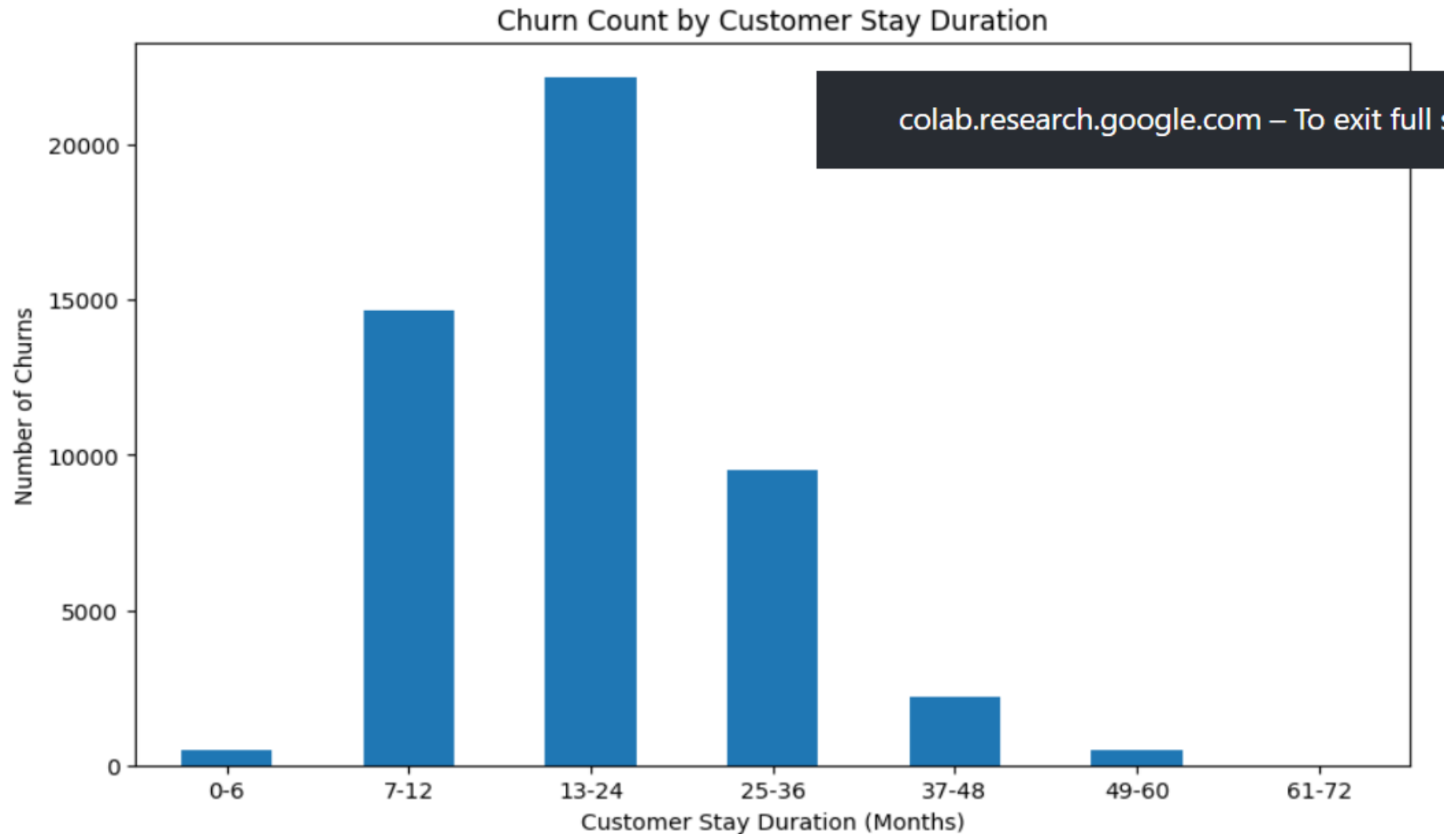
The analysis revealed an inverse relationship between monthly recurring charges and customer churn. Customers subscribed to higher-value plans exhibited lower churn rates, suggesting that premium service offerings, greater customer commitment, and higher perceived value contribute positively to customer retention. Conversely, customers on lower-priced plans were more likely to churn, indicating a need for targeted retention strategies within this segment.

Graphical Evidence Supporting the Second Business Problem (Geographic Variations in Customer Satisfaction and Network Performance)

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7



1

- ❖ Retaining customers for longer periods significantly increases their lifetime value.
- ❖ Customer retention initiatives can directly contribute to revenue growth

2

- ❖ The company should focus on onboarding programs, personalized offers, and upselling opportunities during the first year.
- ❖ Improving engagement early in the customer lifecycle can increase future revenue.

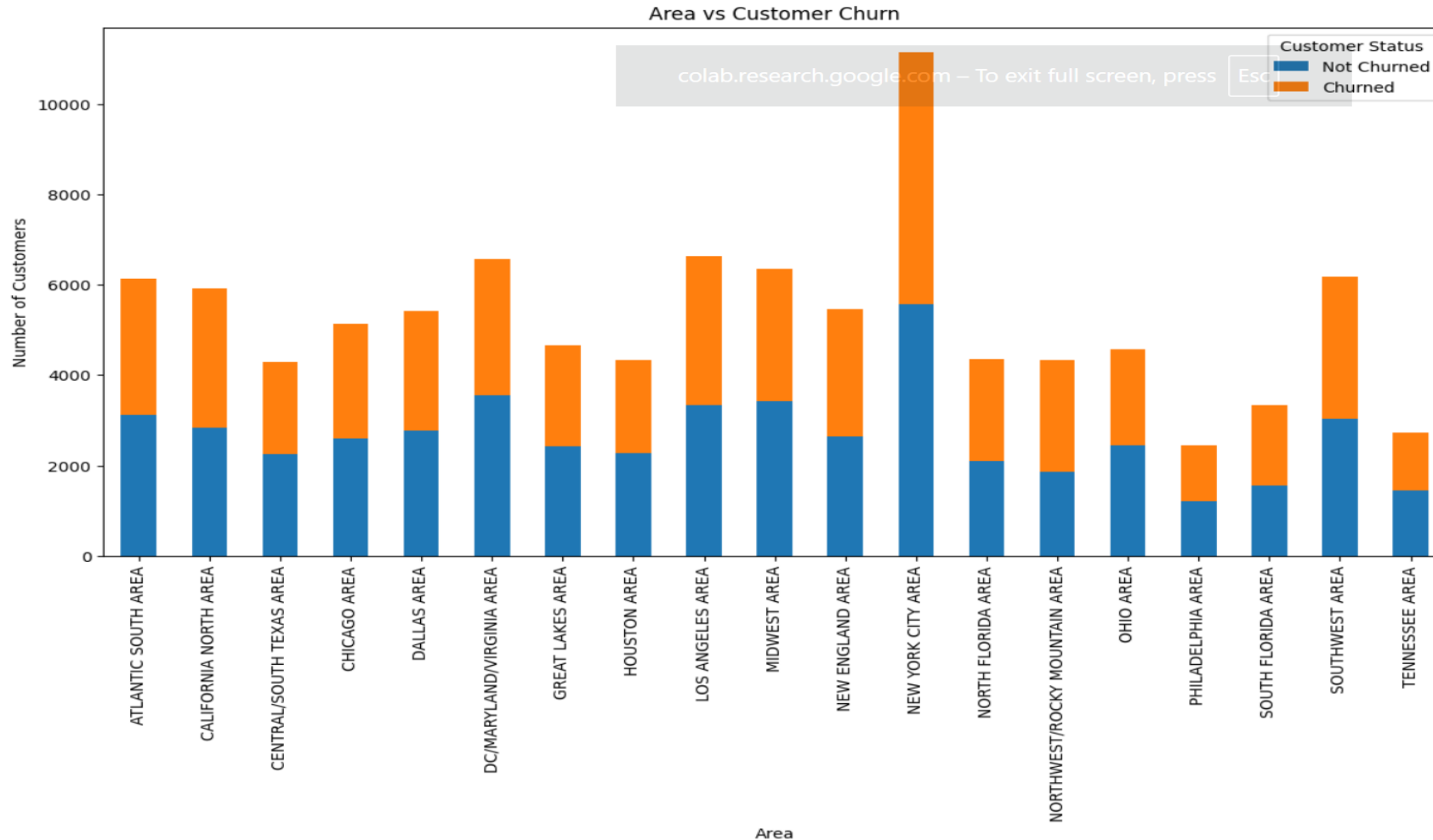
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- ❖ They should be prioritized for loyalty rewards and retention campaigns.
- ❖ Preventing churn in the segment of (48-60) months should be a strategic priority because losing these customers results in significant revenue loss because they have the highest revenue.

4

- ❖ Enhanced customer experience for all subscribers, not only premium subscribers, because Low-value customers are more susceptible to competitor promotions and price-based switching. This will make them put greater financial commitment to the network and be less likely to leave.

Graphical Evidence Supporting the Second Business Problem (Geographic Variations in Customer Satisfaction and Network Performance)



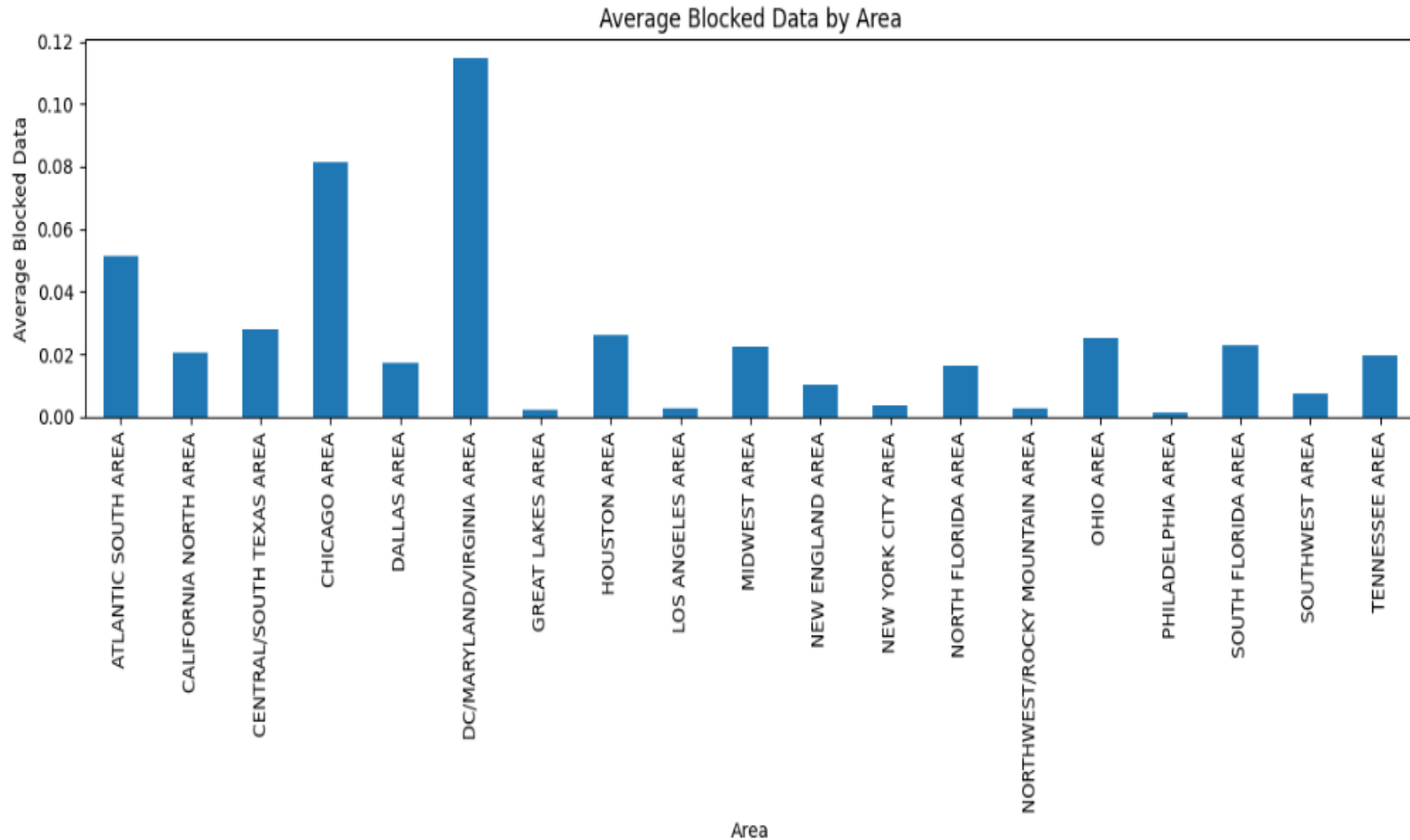
Graphical Evidence Supporting the Second Business Problem (Geographic Variations in Customer Satisfaction and Network Performance)

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1

- ❖ Network expansion
- ❖ Capacity upgrades
- ❖ Infrastructure optimization
- ❖ in the DC/Maryland/Virginia and Chicago regions.

2

- ❖ New England shows the lowest customer experience
- ❖ Recommendation: Conduct customer satisfaction surveys and network quality assessments in these areas.

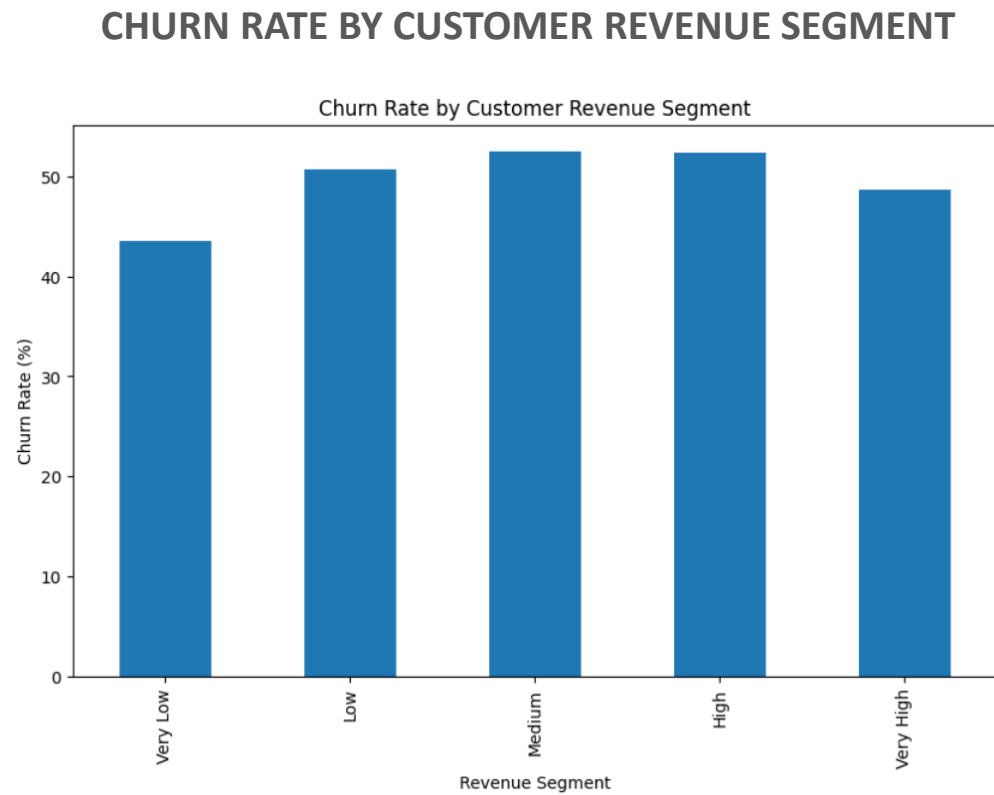
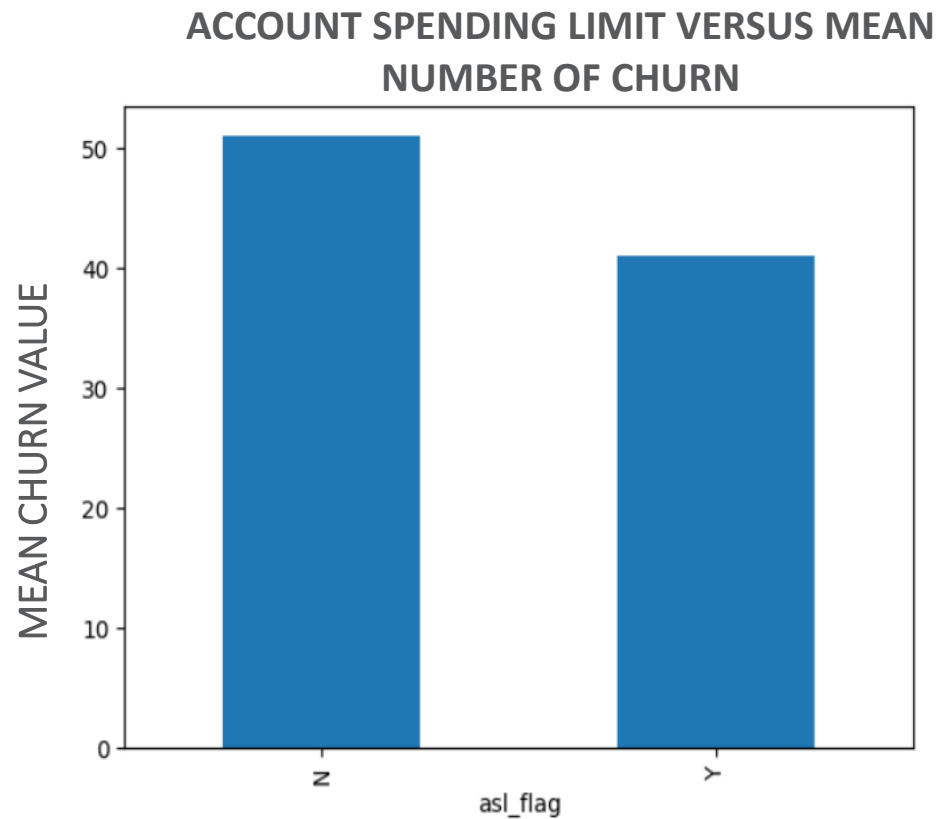
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- ❖ Network quality is important, but customer experience is influenced by multiple factors beyond blocked data alone.

4

- ❖ Enhanced customer experience for all subscribers, not only premium subscribers, because Low-value customers are more susceptible to competitor promotions and price-based switching. This will make them put greater financial commitment to the network and be less likely to leave.

Impact of customer spending behavior and revenue-related factors on customer retention and business profitability.



1

- ❖ **Develop VIP retention programs.**
- ❖ **Offer personalized loyalty incentives.**
- ❖ **Prioritize high-value customers for proactive support.**

2

- ❖ **Focus retention on customers with declining revenue trends.**
- ❖ **Create revenue-based customer segmentation.**

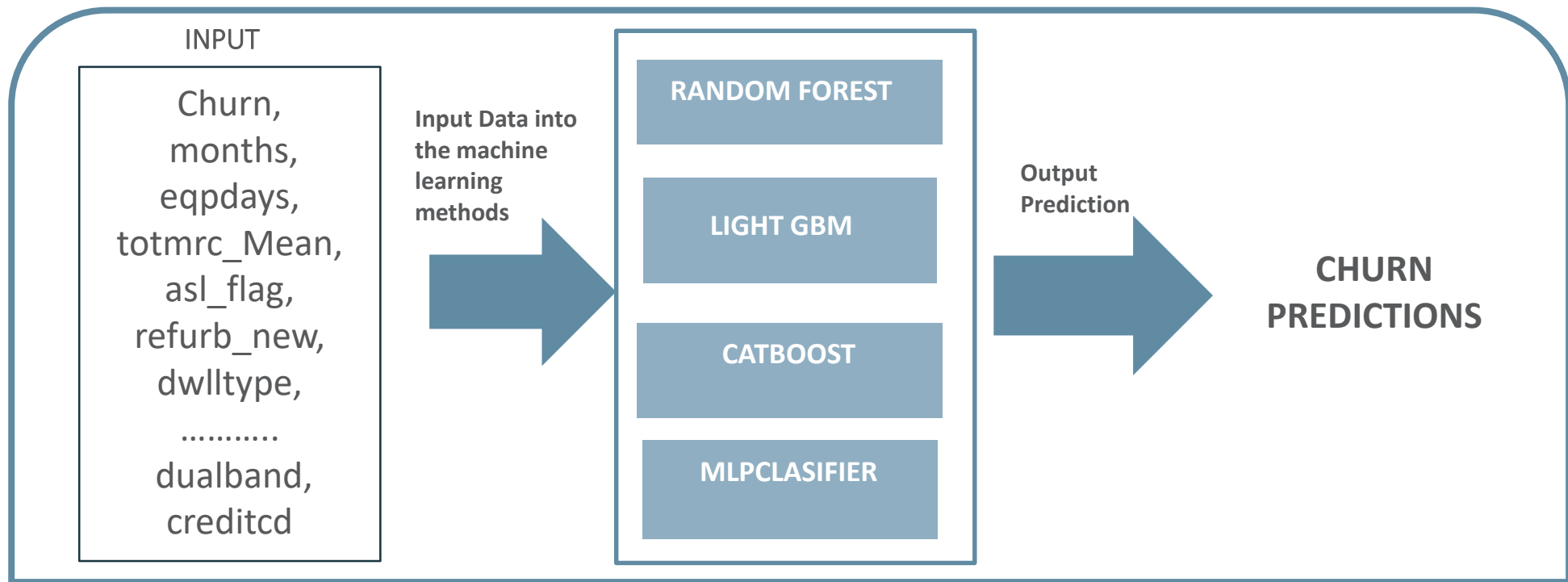
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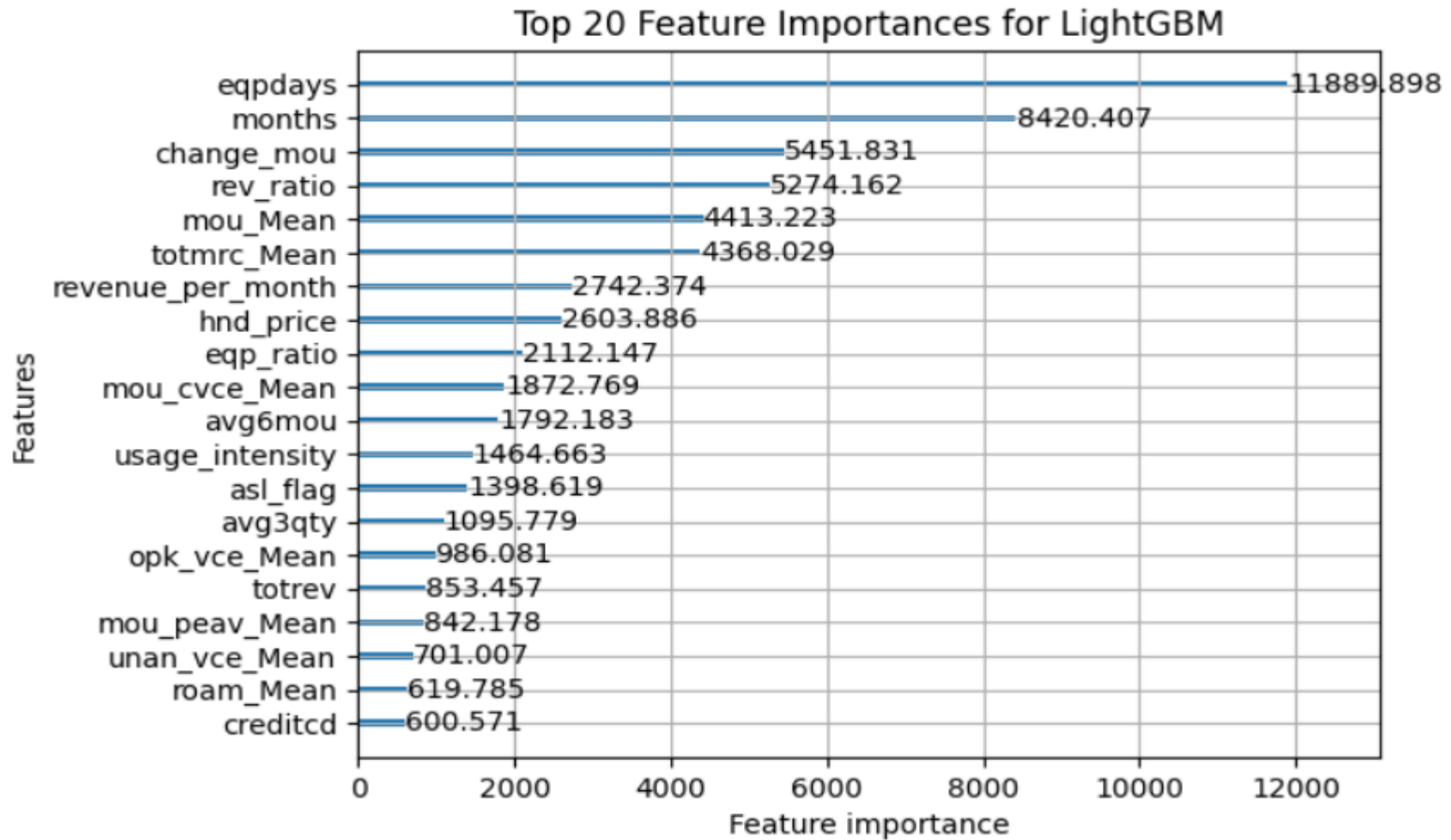
- ❖ **Optimize pricing structures.**
- ❖ **Promote value-added premium packages.**

4

- ❖ **Introduce usage alerts.**
- ❖ **Recommend plan upgrades before limits are exceeded.**
- ❖ **Offer affordable data bundles**

From customer information such as area, months, total revenue, etc., the proposed model predicts the probability





Overview

Using the AI model to predict customer that will churn early

Innovation

Decisions that were once driven by human intuition and experience can now be made faster with greater accuracy

Results of the Analysis on :

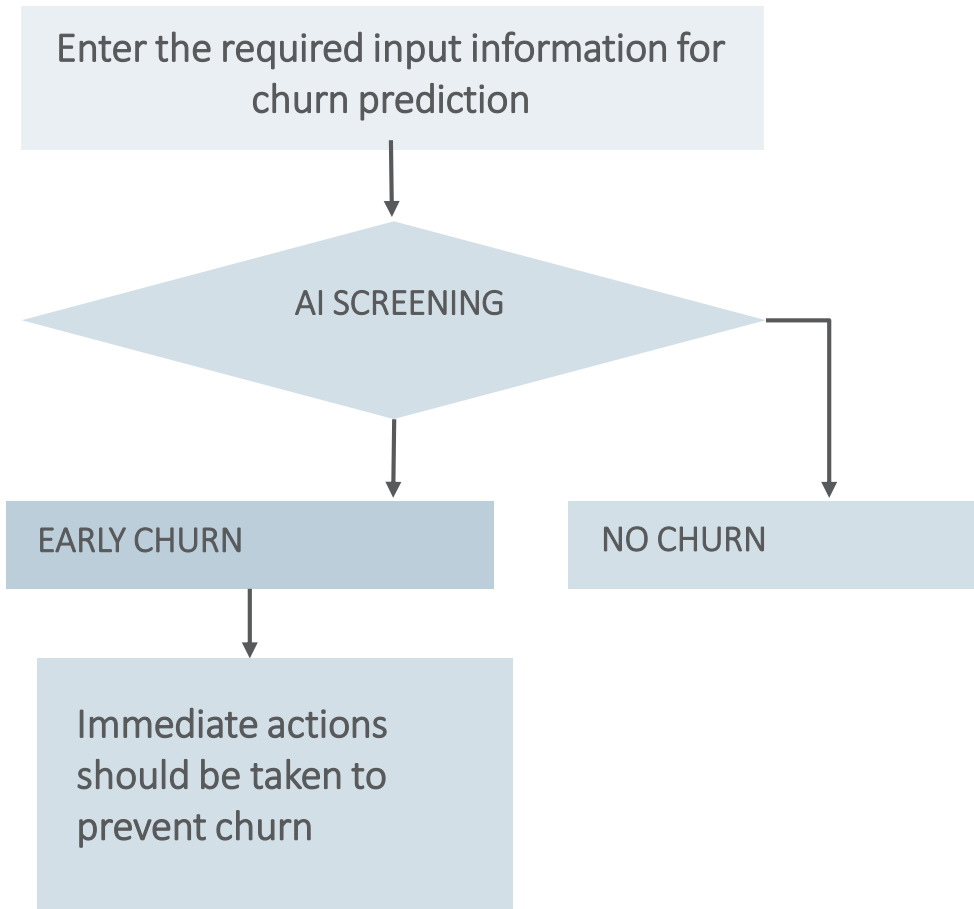
Customers

- ☐ Better customer Service
- ☐ Better network performance
- ☐ Reduced Charges

Company

- ☐ Reduced customer Churn
- ☐ Improve company revenue
- ☐ Improve company productivity

CHURN PREDICTION PROCESS



- ✓ Implement a Proactive Customer Churn Monitoring System
- ✓ Prioritize Retention of High-Value Customers
- ✓ Improve Network Performance in Underperforming Regions
- ✓ Develop Customer-Specific Pricing and Service Packages
- ✓ Manage Data Overage Charges More Effectively
- ✓ Strengthen Customer Engagement During the Early Customer Lifecycle
- ✓ Continuously Monitor Key Churn Drivers

QUOTATION

ACHIEVEMENTS

We developed a machine model that can predict whether a customer will churn or not

Developed key drivers of customer churn

Gave recommendation to the human resource on possible methods to reduce customer churn which will in turn improve the company's productivity

PRICE

**We propose a fee of
500 thousand naira
for providing AI.**

This price corresponds to about 1% of the annual revenue lost by a company in Nigeria

12. Conclusions

The findings of this project demonstrate that customer churn is a significant threat to revenue generation, customer lifetime value, and long-term profitability within the telecommunications industry. The analysis revealed that customer retention is influenced by a combination of factors, including customer tenure, spending behavior, service usage patterns, network performance, and account management practices.

The project successfully demonstrated how business intelligence and machine learning can be integrated to predict customer churn, uncover key drivers of customer attrition, and support strategic decision-making. By enabling the early identification of at-risk customers, the proposed solution offers telecom organizations an effective means of improving customer retention, protecting revenue streams, and enhancing overall business performance.

Chang, V., Hall, K., Xu, Q. A., Amao, F. O., Ganatra, M. A., & Benson, V. (2024). *Prediction of customer churn behavior in the telecommunication industry using machine learning models*. Algorithms, 17(6), 231. <https://doi.org/10.3390/a17060231>

Shaikhsurab, M. A., & Magadum, P. (2024). *Enhancing customer churn prediction in telecommunications: An adaptive ensemble learning approach*. arXiv. <https://arxiv.org/abs/2408.16284>

GSMA Intelligence. (2014). *Mobile churn trends and customer retention strategies: Maintaining market share in the data-centric era*. <https://www.gsmainelligence.com/research/mobile-churn-trends-and-customer-retention-strategies-maintaining-market-share-in-the-data-centric-era>